CSA15 Cloud Computing and Big Data Analytics

Capstone Cloud Technical Presentation

Individual Student Presentation Deck

Presentation Focus

Present your capstone as a cloud-backed product. Explain the problem, architecture, service choices, evidence, and CO1 learning clearly.

Presenter Details

Student Name: B.manujaswanth
Register Number:192525400
Class / Section / Slot: c
Capstone Title: Flood Watch Cloud Early-Warning App Using Rainfall Streams and Community Reports
Selected SDG Goal(s): SDG 9: Industry, Innovation and Infrastructure
SDG 11: Sustainable Cities and Communities
SDG 13: Climate Action
Cloud Platform / Tools: Firebase, Google Cloud Platform (GCP), Firestore

Problem Context and Stakeholder Mapping

Our capstone begins with a real problem to solve

Problem Statement

Real-time flood prediction and early alerts using rainfall data and community reports.

Target Users

Residents in flood-prone areas Disaster management authorities Emergency responders

SDG Fit

SDG 11: Sustainable Cities and Communities **SDG 13:** Climate Action

Product Boundary

Community reporting Flood risk prediction Real-time alerts

Cloud Adoption Rationale and Concept Map

The problem needs cloud support to become a usable product

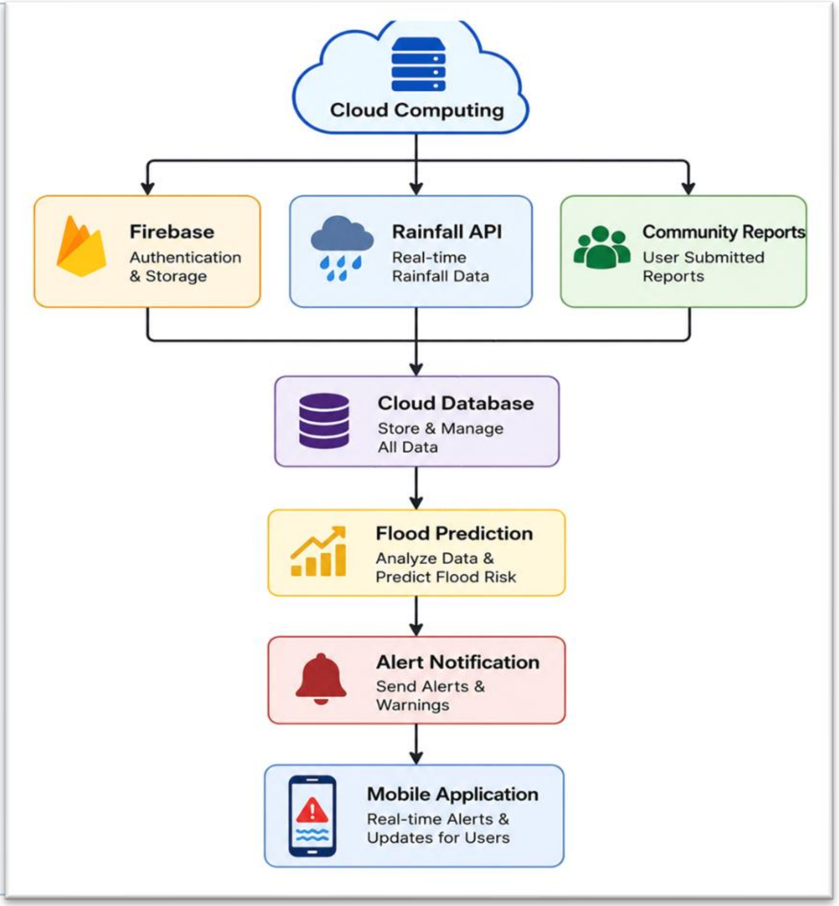
Cloud Definition

Cloud stores, processes, and delivers flood data online.

Cloud Characteristics

Scalability Elasticity Resource Pooling Measured Service

Cloud Concept Map



Capstone Cloud Architecture

Our cloud architecture connects the user action to services

Presentation Points

Scaling point

Flutter Mobile App

Firebase Authentication Rainfall API

Firebase Cloud Functions

Firestore

Security boundary

Python Prediction Model

Cloud Service Model Responsibility Matrix

Each service model defines what we control and what cloud manages

Service area	Model	Provider manages	Student team manages	Why this fits
Frontend hosting	PaaS	Firebase	UI	Easy deployment
Backend/API	FaaS	Firebase	Logic	Serverless
Database	DBaaS	Firestore	Data	Realtime
Storage	SaaS	Firebase	Files	Rules
Security / IAM	IAM	Firebase	Rules	User Authentication

Data Flow and Analytics Pipeline

The product data moves from user activity to useful insight

Rainfall API User Reports — Cloud Functions — Firestore — Python Model — Dashboard Flood Map

Data Requirements

User data, events, logs, files, sensor data, or transactions.

Analytics Requirement

Dashboard, alert, prediction, report, trend, or decision support.

Proof Required

Sample input, sample output, and one insight generated.

Python / AI Module Integration

Python or AI adds intelligence where the product needs it

Python Role

Flood Prediction

Input and Output

Rainfall GPS User Reports, Flood Risk Level

Execution Point

Cloud Function

Product Value

Early Warning

Security, Scalability, Privacy and Cost Design

Security, scale, privacy and cost shape the final design

Security

Firebase Authentication

Privacy

Secure User Data

Scalability

Auto Scaling Cloud

Cost Awareness

Firebase Free Tier.

Implementation Stack and Deployment Evidence

Our implementation stack shows how the design will be built

Mobile / Web Layer

Flutter

Cloud Services

Firebase Firestore Cloud Functions.

Evidence

GitHub Firebase Console.

Version Control

GitHub Repository

Demo Storyboard and User Journey

A complete user journey proves the product flow

Scene 1

User Login

Scene 2

Rainfall Data Collected

Scene 3

Community Report Submitted

Scene 4

Python Predicts Risk

Scene 5

Alert Generated

Scene 6

User Receives Notification

Evaluation Readiness Checklist

Our evidence confirms that the presentation is ready

- [ ] Cloud ecosystem and service model explained
- [ ] Elasticity and scalability decisions are visible
- [ ] REST/SOAP/API role connected to capstone
- [ ] GitHub and cloud evidence are inspectable
- [ ] Architecture has compute, data, storage, security, monitoring
- [ ] References and integrity statement are included

Evaluator Focus

CO1 is verified through cloud design reasoning, service choices, evidence, and individual explanation.

Conclusion, Limitations and Future Scope

We close with learning, limitations and next steps

Learning Outcome

Learned Cloud Computing and Big Data concepts.

Limitations

Limited to selected locations.

Future Improvement

AI improvements and IoT sensor integration.

Integrity Statement

Built using Firebase, Python, Flutter, Rainfall API, Firestore, GitHub, and AI-assisted documentation.